

Prescription Medicine Fields

Frontend Developer Guide

Date: 26 August 2026. Backend is already shipped. Current doctor / nurse / pharmacy screens keep working without frontend changes. New fields are optional. When frontend is ready, send and display them.

Compatibility

Old payload (medicine_name, dosage, frequency, duration, instructions) is still valid. Missing new fields are stored as null and returned as null. Do not wait on a breaking API version.

1. What backend added

Each prescription item (one medicine row) now has extra optional fields. Doctor create/update, consultation save (OPD and IPD), pharmacy detail, and nurse patient medications all return them.

- form - Tablet, Capsule, Syrup, Injection, ...
- dose - amount per take, e.g. 1 or 5 ml (NOT the same as dosage)
- route - Oral, IV, IM, Topical, ...
- timing - After food, Before food, Empty stomach, ...
- quantity - integer total to dispense / take, e.g. 10
- quantity_unit - Tablet(s), Capsule(s), ml, ...

2. UI field vs API key

Map the prescription card to these JSON keys. Do not invent new names.

UI label	API key	Required	Example
Medicine	medicine_name	Yes	Amoxicillin
Strength	dosage	Yes	500 mg
Form	form	No*	Tablet
Dose	dose	No*	1
Route	route	No*	Oral
Frequency	frequency	Yes	1-0-1
Timing / When to take	timing	No	After food
Duration	duration	Yes	5 Days
Quantity number	quantity	No*	10
Quantity unit	quantity_unit	No*	Tablet(s)
Special instructions	instructions	No	Take with water

* Backend does not require the new fields, so today's 6-field doctor form still saves. When you build the new card, treat Form, Dose, Route, Duration and Quantity as required in the UI.

Important: dosage vs dose

dosage = Strength (500 mg). dose = how much to take each time (1 tablet / 5 ml). Do not put both into dosage. Keep them separate in state and in API.

Important: timing vs instructions

timing = After food / Before food. instructions = free-text special notes (max 200 chars in UI). Today the doctor form puts 'After food' in instructions. After the UI change, send timing='After food' and keep instructions for extra notes only.

3. APIs that already include the new fields

Doctor write

- POST /prescriptions body.items[]
- PUT /prescriptions/{id} body.items[]
- OPD consultation save prescription.items[]
- IPD consultation save prescription.items[]

Consultation save already uses PrescriptionItemCreate. No extra nested wrapper. Put the new keys on each item next to medicine_name.

Doctor / pharmacy / nurse read

- GET /prescriptions/{id} items[]
- GET /prescriptions?patient_id= items[]
- Pharmacy prescription detail items[]
- GET /nurse/medications/patient/{patient_id} medications[]

Lab has no prescription API. Lab should not call nurse or pharmacy URLs. If lab later needs medicines, reuse GET /prescriptions with doctor/lab permissions, or skip until product asks for it.

4. Old payload (still works)

```
{
  "appointment_id": 101,
  "diagnosis": "Acute bronchitis",
  "notes": "",
  "items": [
    {
      "medicine_name": "Amoxicillin",
      "dosage": "500 mg",
      "frequency": "1-0-1",
      "duration": "5",
      "instructions": "After food"
    }
  ]
}
```

duration: send number or '5 Days'. Backend stores text like '5 days'. Empty new fields are omitted; backend stores null.

5. New payload (when frontend is ready)

```
{
  "appointment_id": 101,
  "diagnosis": "Acute bronchitis",
  "notes": "",
  "items": [
    {
      "medicine_name": "Amoxicillin",
      "dosage": "500 mg",
      "form": "Tablet",
      "dose": "1",
      "route": "Oral",
      "frequency": "1-0-1",
      "timing": "After food",
      "duration": "5 Days",
    }
  ]
}
```

```

    "quantity": 10,
    "quantity_unit": "Tablet(s)",
    "instructions": "Take with water, avoid cold drinks"
  }
]
}

```

IPD: send admission_id instead of appointment_id (exactly one parent). quantity must be an integer >= 1, or omit it. Do not send "". Blank strings for form/dose/route/timing are stored as null.

6. Response shape (item)

```

{
  "id": 55,
  "medicine_name": "Amoxicillin",
  "dosage": "500 mg",
  "form": "Tablet",
  "dose": "1",
  "route": "Oral",
  "frequency": "1-0-1",
  "timing": "After food",
  "duration": "5 Days",
  "quantity": 10,
  "quantity_unit": "Tablet(s)",
  "instructions": "Take with water, avoid cold drinks"
}

```

Old rows: form, dose, route, timing, quantity, quantity_unit are null. UI must render '-' / empty, not crash.

7. Suggested doctor UI state

```

{
  name: "",
  dosage: "",
  form: "Tablet",
  dose: "",
  route: "Oral",
  frequency: "1-0-1",
  timing: "After food",
  durationValue: "",
  durationUnit: "Days",
  quantity: "",
  quantityUnit: "Tablet(s)",
  instructions: ""
}

```

Mapper UI -> API

- name -> medicine_name
- dosage -> dosage (Strength, e.g. 500 mg)
- form, dose, route, frequency, timing as-is
- durationValue + durationUnit -> duration string, e.g. '5 Days'
- quantity as Number; omit if empty
- quantityUnit -> quantity_unit
- instructions as-is (special notes only)

Mapper API -> UI

- Parse duration '5 Days' back to durationValue=5, durationUnit=Days
- Null new fields -> empty string in form state
- Keep existing medicines[] shape; add the extra keys on each row

8. Suggested dropdown values

- Form: Tablet, Capsule, Syrup, Injection, Drops, Cream, Ointment, Inhaler
- Route: Oral, IV, IM, SC, Topical, Inhalation, Sublingual
- Frequency: 1-0-1, 1-1-1, 1-0-0, 0-0-1, 1-1-0, SOS, STAT
- Timing: After food, Before food, Empty stomach, With food, At bedtime
- Duration unit: Days, Weeks, Months
- Quantity unit: Tablet(s), Capsule(s), ml, Bottle(s), Tube(s)

These are UI lists only. Backend stores free text (max 50 chars). Medicine name can stay a text input until a medicine catalog exists.

9. Doctor screens to update later

- ConsultationModal Prescription tab (main card from the screenshot)
- PrescriptionDetailModal (view / edit / add medicine)
- QuickPrescribeModal
- clinicalMapper.js uiMedicinesToApiItems and apiToUiPrescription
- DEFAULT_MEDICINE in doctor/constants.js
- Patient history / prescription print if it lists medicines

Also add: per-card delete, + Add medicine, info text 'Prescription is saved when you click Save Consultation.', and 0/200 counter on special instructions.

10. Nurse (read-only display)

GET /nurse/medications/patient/{patient_id} medications[] now includes dosage, dose, form, route, timing, quantity, quantity_unit, instructions.

Current nurse mapper

Today nurseMapper maps dose from dosage first (item.dosage ?? item.dose). After backend returns both, Dosage column still shows Strength until you change the mapper to: dose: item.dose, and keep dosage/strength separate. Route already prefers item.route, then instructions. Once doctors send route, Route column shows Oral instead of After food. That is correct.

Recommended nurse mapping

```
{
  medicine_name: item.medicine_name,
  strength: item.dosage,           // 500 mg
  dose: item.dose,                 // 1
  form: item.form,
  route: item.route,
  timing: item.timing,
  frequency: item.frequency,
  duration: item.duration,
  quantity: item.quantity,
  quantity_unit: item.quantity_unit,
  instructions: item.instructions
}
```

}

- Do not let nurses create or edit prescription fields
- Null-safe: show '-' when a field is null (old prescriptions)
- Optional extra columns: Form, Timing, Quantity

11. Pharmacy

Pharmacy detail items[] now include form, dose, route, timing, quantity, quantity_unit. quantity_prescribed uses item.quantity when it is a positive integer. Old prescriptions still use duration as supply count.

- Show Strength from dosage, Dose from dose, Form, Route, Timing
- Total quantity: prefer quantity + quantity_unit when present
- quantity_prescribed / quantity_dispensed / quantity_remaining unchanged
- Do not write prescription fields; pharmacist only dispenses

12. Lab

No backend change for lab. Lab orders stay on the Lab and Follow-up tab. Do not mix lab test payload with prescription items. If product later wants lab to see medicines, read the same prescription GET items[] (read-only): medicine_name, dose, route, timing, instructions.

13. Validation rules (frontend)

- If medicine_name is filled, require duration > 0 (already in UI)
- When new card is live: also require dosage, form, dose, route, frequency, quantity
- quantity integer >= 1; omit key if the row is unused
- instructions max 200 characters
- Do not send duration as 0; backend still requires duration text
- Empty optional strings are OK; backend converts them to null

14. Files frontend will change (later)

- src/features/doctor/constants.js
- src/features/doctor/components/ConsultationModal.jsx
- src/features/doctor/components/PrescriptionDetailModal.jsx
- src/features/doctor/components/QuickPrescribeModal.jsx
- src/shared/api/mappers/clinicalMapper.js
- src/shared/api/mappers/nurseMapper.js (dose/route mapping)
- Nurse medication tables / patient overview
- Pharmacy prescription detail / dispense table (optional columns)

15. QA checklist

- OLD UI: save consultation with only 6 fields still succeeds
- OLD UI: reload prescription; existing 6 fields still populated
- NEW UI: save all extra fields; GET returns the same values
- NEW UI: omit timing/instructions; they come back as null
- Nurse: old Rx still lists medicine/dosage/frequency
- Nurse: new Rx can show route/dose when mapper is updated
- Pharmacy: old Rx quantity_prescribed still derived from duration

- Pharmacy: new Rx with quantity=10 uses 10 as prescribed supply
- IPD consult save with admission_id still works
- Add second medicine; both items persist extra fields

16. Backend note for local runs

```
cd HM-System-Backend  
alembic upgrade head
```

Migration c2d3e4f5g6h7 adds nullable columns on prescription_items. Until it is applied, SQLAlchemy will error on the new fields. No permission or seed change. No frontend deploy is required for current screens to keep working after the migration.